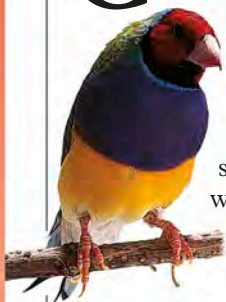


COLOR



Imagine waking up in a world without color. There would be no beautiful paintings or rainbows to look at, and no bright shoes or clothes to wear. At night when you switch off the light, all the bright colors around you suddenly disappear. This is

because you cannot see the color of an object without light. White light can be split into all the colors of the rainbow—shades of red, orange, yellow, green, blue, indigo, and violet. When all these colors are mixed back together, they make white light again.

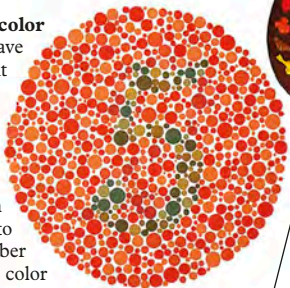


Seeing red

We see things because light bounces off them. Light is actually a mixture of all the colors of the rainbow. When light hits these shoes, all the colors sink in, except the red color that is reflected back to your eyes.

Seeing in color

If you don't have all the different types of cones in your eyes, you may find it hard to tell some colors apart. A color-blind person may not be able to see the number on this color blindness test.

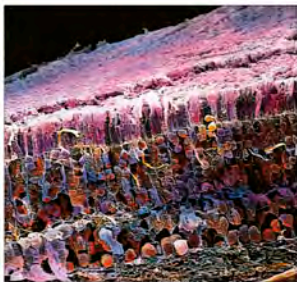


White is the only color that can't be made by mixing together the primary colors of paint.



Light mixing

The colors of light behave differently than paint colors. When red, green, and blue light are mixed together, they make white light. When the same colors of paint are mixed, it makes a dark color.



Color cones

The inside of the back of your eye looks like this through a microscope. You have cells in the back of your eyes called cones, which send messages to your brain about the colors you see.